

Practical No. 8

Aim: To perform data aggregation using MongoDB aggregation pipeline.

Output:

```
> db.students.aggregate([{$group: {_id: "$department", count: {$sum: 1}}]])
< {
  _id: 'IT',
  count: 1
}
{
  _id: 'DS',
  count: 2
}
{
  _id: 'AI',
  count: 2
}
{
  _id: 'CS',
  count: 3
}
```

```
> db.students.aggregate([{$sort: {marks: -1}}])
< {
  _id: ObjectId('69a9cfe45200ba3b052b9210'),
  name: 'Elon Musk',
  department: 'AI',
  age: 22,
  email: 'elonmusk@gmail.com',
  marks: 91
}
{
  _id: ObjectId('69a9cfe45200ba3b052b920f'),
  name: 'Sundar Pichai',
  department: 'CS',
  age: 21,
  email: 'sundarpichai@gmail.com',
  marks: 90
}
{
  _id: ObjectId('69a9cfe45200ba3b052b9211'),
  name: 'Bill Gates',
  department: 'CS',
  age: 23,
  email: 'billgates@gmail.com',
  marks: 88
}
{
  _id: ObjectId('69a9cfe45200ba3b052b9214'),
  name: 'Sergey Brin',
  department: 'CS',
  age: 21,
  email: 'sergeybrin@gmail.com',
  marks: 87
}
{
  _id: ObjectId('69a9cfe45200ba3b052b920d'),
  name: 'Sam Altman',
  department: 'AI',
  age: 21,
  email: 'samaltman@gmail.com',
  marks: 85
}
```

```
> db.students.aggregate([{$match: {marks: {$gt: 85}}]])
< {
  _id: ObjectId('69a9cfe45200ba3b052b920f'),
  name: 'Sundar Pichai',
  department: 'CS',
  age: 21,
  email: 'sundarpichai@gmail.com',
  marks: 90
}
{
  _id: ObjectId('69a9cfe45200ba3b052b9210'),
  name: 'Elon Musk',
  department: 'AI',
  age: 22,
  email: 'elonmusk@gmail.com',
  marks: 91
}
{
  _id: ObjectId('69a9cfe45200ba3b052b9211'),
  name: 'Bill Gates',
  department: 'CS',
  age: 23,
  email: 'billgates@gmail.com',
  marks: 88
}
{
  _id: ObjectId('69a9cfe45200ba3b052b9214'),
  name: 'Sergey Brin',
  department: 'CS',
  age: 21,
  email: 'sergeybrin@gmail.com',
  marks: 87
}
```

```
> db.students.aggregate([{$project: {name: 1, marks: 1, _id: 0}}])
< {
  name: 'Sam Altman',
  marks: 85
}
{
  name: 'Jeff Bezos',
  marks: 78
}
{
  name: 'Sundar Pichai',
  marks: 90
}
{
  name: 'Elon Musk',
  marks: 91
}
{
  name: 'Bill Gates',
  marks: 88
}
```

```
> db.students.aggregate([{$lookup: {from: "dept", localField: "department", foreignField: "dept", as: "department_info"}}])
< {
  _id: ObjectId('69a9cfe45200ba3b052b920d'),
  name: 'Sam Altman',
  department: 'AI',
  age: 21,
  email: 'samaltman@gmail.com',
  marks: 85,
  department_info: [
    {
      _id: ObjectId('69bd47874d5392a3bb4a8c57'),
      dept: 'AI',
      hod: 'Levi Ackerman'
    }
  ]
}
{
  _id: ObjectId('69a9cfe45200ba3b052b920e'),
  name: 'Jeff Bezos',
  department: 'DS',
  age: 20,
  email: 'jeffbezos@gmail.com',
  marks: 78,
  department_info: [
    {
      _id: ObjectId('69bd47874d5392a3bb4a8c58'),
      dept: 'DS',
      hod: 'Light Yagami'
    }
  ]
}
}
```

```
> db.students.aggregate([{$limit: 2}])
< {
  _id: ObjectId('69a9cfe45200ba3b052b920d'),
  name: 'Sam Altman',
  department: 'AI',
  age: 21,
  email: 'samaltman@gmail.com',
  marks: 85
}
{
  _id: ObjectId('69a9cfe45200ba3b052b920e'),
  name: 'Jeff Bezos',
  department: 'DS',
  age: 20,
  email: 'jeffbezos@gmail.com',
  marks: 78
}
}
```

```
> db.students.aggregate([{$match: {marks: {$gt: 85}}}, {$project: {name: 1, marks: 1, _id: 0}}, {$sort: {marks: -1}}, {$limit: 3}])
< {
  name: 'Elon Musk',
  marks: 91
}
{
  name: 'Sundar Pichai',
  marks: 90
}
{
  name: 'Bill Gates',
  marks: 88
}
}
```

Commands:

```
db.students.aggregate([{$group: {_id: "$department", count: {$sum: 1}}]])
```

```
db.students.aggregate([{$match: {marks: {$gt: 85}}]])
```

```
db.students.aggregate([{$sort: {marks: -1}}]])
```

```
db.students.aggregate([{$project: {name: 1, marks: 1, _id: 0}}]])
```

```
db. students. aggregate([{$lookup: {from: "dept", localField: "department", foreignField: "dept", as: "department_info"}}])
```

```
db.students.aggregate([{$limit: 2}])
```

```
db.students.aggregate([{$match: {marks: {$gt: 85}}}, {$project: {name: 1, marks: 1, _id: 0}}, {$sort: {marks: -1}}, {$limit: 3}])
```